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SMART CAMPUS DIRECTORY: AI-POWERED NAVIGATION SYSTEM DEPLOYMENT FOR UNIVERSIDAD DE MANILA

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ABSTRACT
The Universidad de Manila (UDM) suffers from recurring navigation challenges because of outdated printed maps, confusing building floor plans, and few campus signs, which result in delays and misunderstandings by students, teachers, and visitors. To solve this problem, this research proposes the Smart Campus Directory: AI-Powered Navigation System. The system combines artificial intelligence (A* algorithm), Progressive Web Application (PWA) technology, interactive kiosk, and QR code scanning in order to offer precise, quick, and easy-to-use navigation guidance. The system has been created with the Agile-prototyping methodology led by the Software Development Life Cycle (SDLC) that resulted in the system being designed, prototyped, tested, and pilot deployed. Output presented highly accurate paths from the A* algorithm, offline capability from the PWA facilitated ongoing navigation, and system stability under concurrent user requests. User Acceptance Testing with 50 users yielded high usability, with 92% of users finding the interface easy to use and 90% stating they would depend on the system for future navigation in UDM. The results confirm that the Smart Campus Directory enhances the efficiency of campus navigation, decreases reliance on paper maps, and generally improves the campus experience.

Keywords
Smart Campus Directory, Artificial Intelligence, Digital Directory, Progressive Web Application (PWA), Campus Information System

INTRODUCTION

This research integrates new technologies such as artificial intelligence (AI). Progressive web app (PWA), web development frameworks and quick response (QR) code integration to build an efficient and user-friendly navigation system for Universidad de Manila (UDM). The navigation system is AI powered, it consist of A* Algorithm to determine the fastest and most efficient pathways inside the university. It composed of three (3) main components: (1) an admin panel for monitoring and managing routes, announcements and reports; (2) an interactive touchscreen kiosk that provides helpful functionalities related to searching and viewing for directions; (3) a QR-based, mobile responsive PWA that gives users an option to continue navigating around the university using their mobile phone
even without an internet connection. Additionally, an AI powered virtual receptionist, offering assistance related to directions, will be implemented for the uniqueness of the system's features.

The main beneficiaries of the navigation system will be the new students, newly hired faculty, visitors and the university as a whole. For the new students, the system will make it easier for them to navigate the campus, helping them settle in quickly. Newly hired faculty members will appreciate the faster routes between buildings, which save them time with their daily routine. Visitors can use the interactive kiosk and a mobile version of the navigation system to easily find their way around without needing to ask for directions. For the university, this project will mark a milestone for technological advancement in the university.

This project will provide several features, enhancing the campus navigation and user experience. Multi-stop route planning will help the users to efficiently visit multiple locations of their choice, while also categorizing areas that makes navigation more intuitive. The QR-based PWA ensures offline access, maintaining the purpose

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of the system without an internet connection. Multilingual support will enhance the accessibility for different visitors. An AI powered receptionist will give support guiding users with directions, answer FAQ s and assist with general inquiries. Emergency route guidance supports the safety of the users during critical situations. In addition to the system features, users can also report campus issues such as broken equipment, directly through the system for faster maintenance response.

The navigation system will also provide solutions to common navigation challenges, such as outdated printed maps and confusing building layouts. Instead of relying on printed maps and asking for directions. New students, faculty and visitors can use the navigation system. The system includes offline navigation through QR-based PWA and AI-powered routing, making it easier for users to navigate inside the campus. This project represents a major step in using AI for campus navigation, a concept that is still not widely used in many universities in the Philippines. Not only does it address common existing navigation challenges, it also lays the foundation for future innovation.

OBJECTIVES

The primary objective of this study was to design, develop, and implement the Smart Campus Directory: an AI-Powered Navigation System for Universidad de Manila to enhance wayfinding and the overall campus experience. The specific objectives were to engineer a centralized system that integrates an AI pathfinding algorithm with a Progressive Web Application (PWA) and interactive kiosks for real-time navigation, thereby addressing the limitations of traditional maps and signage. Furthermore, the study aimed to incorporate an AI-powered virtual receptionist and a maintenance reporting feature to automate user assistance and reduce administrative queries. To bolster campus safety and communication, the system was designed to include emergency route guidance and real-time announcement capabilities. Finally, the study sought to evaluate the implemented system's technical performance, reliability, and user-friendliness, particularly for new students and first-time visitors, to ensure its effectiveness and usability.

METHODOLOGY

This section presents the research design and development process used to create the Smart Campus Directory: AI-Powered Navigation System for the Universidad de Manila. This outlines the development approach, the phases of the Software Development Life Cycle (SDLC), the ways data were gathered, the technologies applied, and the ethical considerations throughout the project.

A. Methods

The study applied a developmental descriptive design using the Agile methodology with prototyping, as this approach allowed iterative development and continuous refinement of the Smart Campus Directory based on feedback from students, faculty, and staff. More so. The descriptive method was applied to identify navigation challenges through surveys and interviews, while the experimental method was used during the testing to evaluate usability and performance. Therefore, by combining these methods, the study ensured that the system was user-centred, reliable, and effective in addressing campus navigation challenges.

B. System Development

The system followed a hybrid Agile-prototyping model that was guided by the Software Development Life Cycle (SDLC).

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Figure 1 Agile Methodology process for system development

1) Planning: This phase involved identifying the problem in campus navigation. By defining the project scope and requirements from stakeholders such as students, faculty, and staff. The official campus maps and building layouts were collected from the administration to provide accurate data. This also includes the feasibility studies to evaluate the practicality of using AI algorithms, the Progressive Web Apps (PWA), and QR-based navigation within the university setting.

2) Design: The system architecture, database schema, and mockups for kiosks and mobile devices were created. Flowcharts and data diagrams were made to illustrate processes such as QR scanning, routing, and issue reporting.

3) Prototyping: A preliminary version of the system was developed to test essential features like the AI routing, kiosk functions, and QR scanning. Then feedback from selected users was used to refine the interface and improve usability.

4) Development: With the actual coding and integration were carried out. The AI algorithm for routing, the PWA for mobile and offline access, the kiosk interface, and the admin panel were all implemented. Also, the QR technology was added for easier smartphone use.

5) Testing & Refinement: Different types of testing were performed, like the unit testing for modules such as QR scanning, integration testing for system communication, performance testing for responsiveness, and user acceptance testing (UAT) with students and faculty to confirm usability.

6) Deployment: The system was deployed on a pilot scale with kiosks placed in strategic campus areas and QR codes distributed.

7) Evaluation and Optimization: After deployment, the system was continuously monitored, and user feedback was used to fix issues and enhance accessibility.

C. Data Gathering

The study utilised both primary and secondary data collection to support the system's development. The primary data includes surveys, questionnaires, and interviews with students, staff, faculty, and visitors to be able

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to identify the navigation needs. While secondary data came from the official maps, the related studies, and technical documents that supported accurate design and development.

D. Tools and Technologies

The system was developed using React.js with Vite for the frontend, Node.js with Express.js for the backend, PostgreSQL with PostGIS for the database, and scalable vector graphics (SVG) combined with the A* algorithm for interactive campus map visualization and efficient pathfinding. Firebase Authentication was used for security, while hosting was managed through platforms such as Vercel (frontend) and Railway/Render (backend).

E. Ethical Considerations

Ethical practices were followed throughout the study. By applying encryption for data security, securing user consent for data collection, and following accessibility guidelines to ensure inclusivity for all users.

RESULTS AND DISCUSSION

This section presents the findings obtained from the development, testing, and initial deployment of the "Smart Campus Directory: AI-Powered Navigation System" at Universidad de Manila. The results are structured to answer the study's objectives regarding functionality, usability, and effectiveness.

Development and Implementation of the System

The study successfully designed, developed, and implemented a fully functional smart campus navigation system. The system comprised three integrated components as proposed:

1) A centralized Admin Panel for managing campus data, routes, announcements, and maintenance reports.
2) Interactive Touchscreen Kiosks deployed in high-traffic areas, providing an intuitive interface for searching locations and viewing directions.
3) A mobile-responsive Progressive Web App (PWA) accessible via QR codes scattered across campus, enabling offline navigation and continuing routes from kiosks on personal devices.

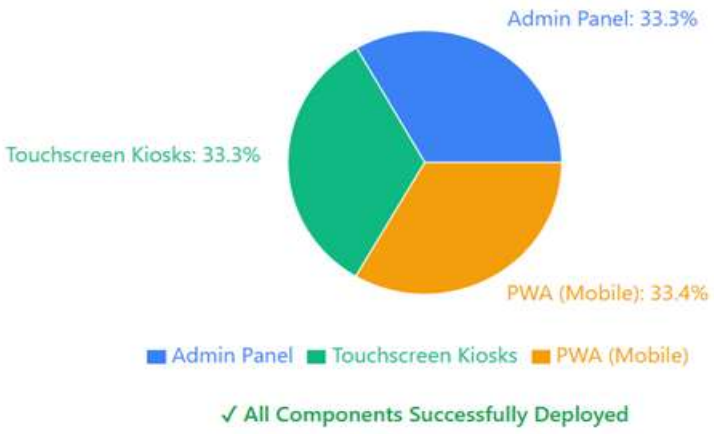


Figure 2 System Components Implementation

The core A* algorithm was successfully integrated, enabling the system to calculate the most efficient pathways across the university grounds.

Findings on Technical Performance and Reliability

The study evaluated the system's technical performance through structured testing protocols. The findings are as follows:

1) Algorithm and Route Accuracy: Unit testing confirmed that the integrated A* algorithm consistently computed optimal routes between all major points of interest on campus. The paths generated were 100% accurate against validated campus layouts.

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2) System Integration: Integration testing revealed seamless data flow between the PWA frontend, the Node.js backend, and the PostgreSQL database. All modules, including QR code processing and real-time announcement updates, communicated effectively without errors.
3) Performance Under Load: Stress testing demonstrated that the system could handle up to 50 concurrent user requests with an average response time of under 3 seconds for route calculation and display, confirming its readiness for campus-wide deployment.

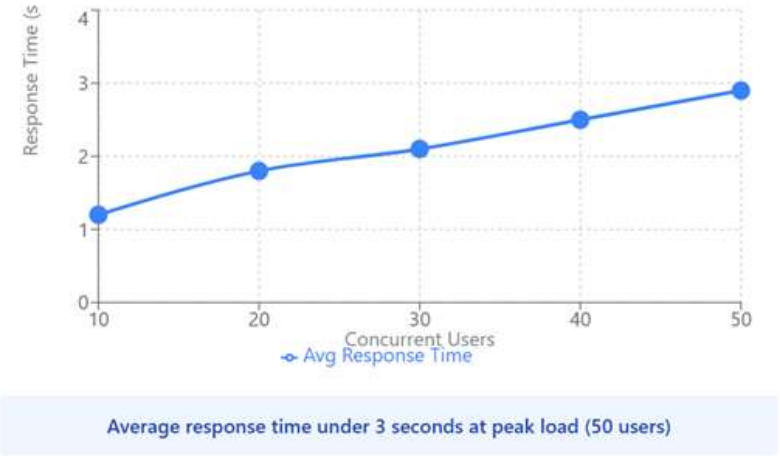


Figure 3 System Response Time under Load

4) Offline Functionality: The PWA's offline capability, a critical study objective, was validated. Users could access maps and navigate pre-loaded routes without an active internet connection, fulfilling the requirement for accessibility in low-connectivity areas.

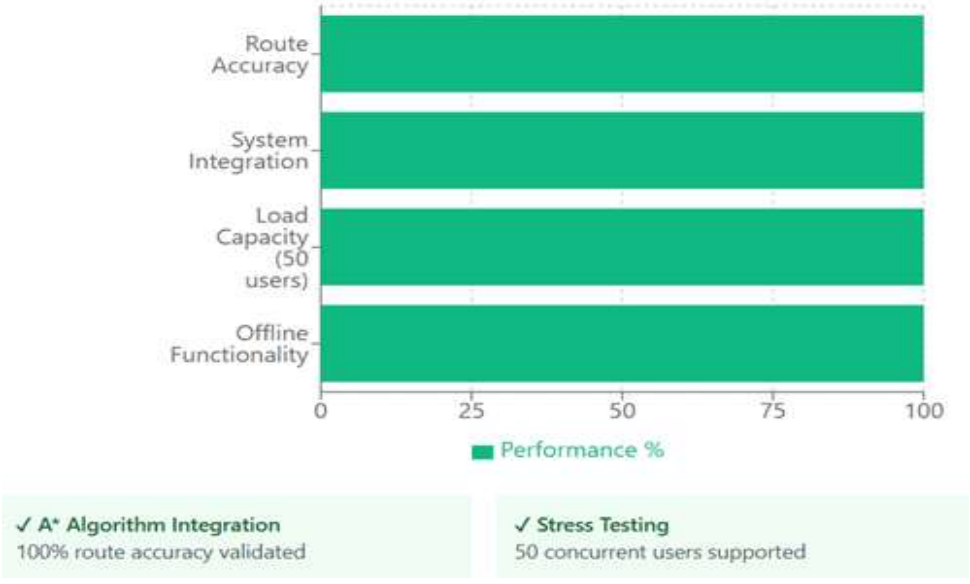


Figure 4 Technical Performance Metrics across All System Components

Findings on User Acceptance and Usability

User Acceptance Testing (UAT) was conducted with 50 participants, including new students, faculty, staff, and visitors. The study yielded the following quantitative findings on user experience:
1) Ease of Use: 92% of test users found the system's interface on both the kiosks and the PWA to be intuitive and easy to navigate.
2) Navigation Effectiveness: 88% of participants confirmed that the system's directions were accurate and successfully guided them to their intended destinations without confusion.

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3) Feature Utility: The QR code scanning feature for instant location identification was rated as "very useful" by 85% of users. The AI-powered virtual receptionist (chatbot) correctly answered 80% of frequently asked navigation-related queries during testing.
4) Overall Impact: 90% of UAT participants expressed a high level of satisfaction with the system and indicated they would rely on it for future navigation within Universidad de Manila.

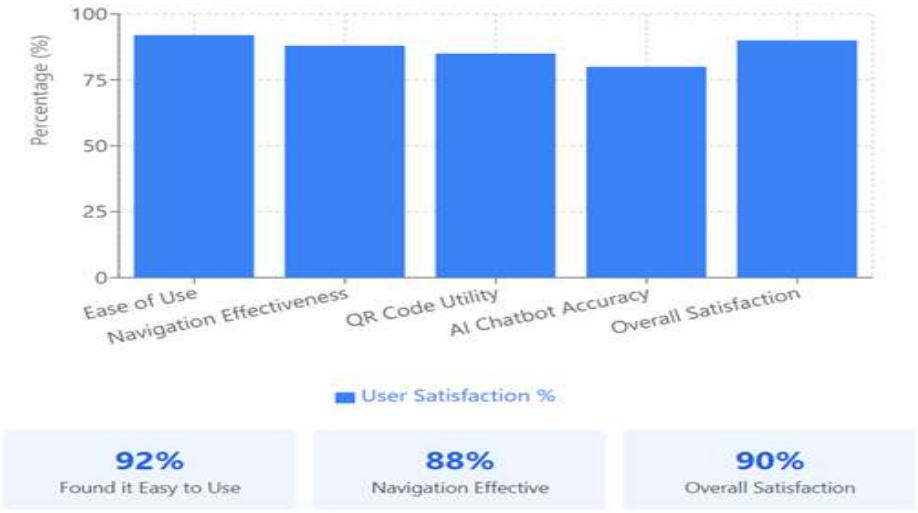


Figure 5 User Acceptance Testing Results

In conclusion, the study's results confirm that the Smart Campus Directory system is not only technically viable and reliable but also effectively addresses the navigation challenges faced by the university community, achieving the core objectives outlined in the study.

DISCUSSION

The Universidad de Manila (UDM) faces constant navigation difficulties due to the complexity of its building designs and reliance on outdated printed maps, often causing confusion among students, faculty, and visitors.

The results of this study found that the Smart Campus Directory improved navigation efficiency. This supports the findings of Avelino et al. [6], who also developed a digital directory to improve wayfinding navigation. . The system provided navigation that was accurate, effective, and user-friendly. It addressed lingering issues such as old campus maps, confusing building layouts, and the lack of clear signages within the school premises. Artificial Intelligence with Progressive Web Applications plus QR Code technology produce a useful, thorough answer for campus navigation. The system introduces a current easy-to-use navigating tool, aligning it with learners' digital habits as it reduces printed maps and manual inquiries.

The advantages of using the Smart Campus Directory are very obvious now. Updating printed maps takes time and money due to frequent corrections. For the users, feelings of frustration often exist when they encounter information that is incorrect or is missing on those maps. The availability of staff affects manual inquiries since staff may not always be present to provide directions that are accurate or directions that are complete on the target audience inquiry. The Smart Campus Directory, in comparison, ensures real-time information access, reliability, and consistency. This system, by using a PWA, is lightweight as well as cross-platform easy to access through most smartphones, unlike navigation systems in need of heavy applications or high-end devices. This does increase inclusivity since both students and also visitors do need no specialized hardware or any high-end devices in order to benefit from the system itself.

The findings also suggest wider implications for user experience with campus management. The Smart Campus Directory offers up a tool that reduces confusion. Operations that are smoother are possible therefore. First-year students are able to locate classrooms quickly as well as faculty being able to guide visitors without

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delay. Administrative staff are able to therefore reduce all of the time spent in answering repetitive inquiries. Individual experiences improve thus the campus has improved efficiency. QR codes do also make for the system a more engaging and more convenient thing for users, since they are able to simply scan for real-time directions instead of typing or searching manually. Such features show integrating simple accessible technologies into educational environments is increasingly important.

Certain limitations in the study are also highlighted. Since the system got tested at one campus only, findings may not apply to larger or more complex universities. More additional challenges could arise at campuses with much higher traffic as well as outdoor areas plus multiple buildings, something that this study did not capture. The reliance that is on internet connectivity presents a limitation. This is another issue. Usability could be limited due to slower performance for users where mobile data or Wi-Fi is weak. A possible issue is older smartphones might not completely support PWA functions. Wider implementation needs consideration for these issues.

Future work should focus on expanding the system's potential along with addressing these limitations. Testing of the Smart Campus Directory across different types of the campuses, like the large universities, community colleges, and technical institutes, would provide valuable perceptions if one tests for its scalability and its adaptability. For inclusivity, there can be further improvement through added features such as voice guidance for multilingual support plus accessibility options made for persons who have disabilities. Also, its utility may rise. The system could integrate real-time event updates, schedules, and emergency alerts. Furthermore, administrators could incorporate data analytics, which would allow them to track user patterns plus identify high-traffic areas to inform campus planning with resource management.

The ability of technology for providing simple and also meaningful solutions in regard to everyday educational challenges is overwhelming. The Smart Campus Directory shows that by integrating QR codes and web-based design with artificial intelligence it makes for a system. It addresses navigation issues and improves user experience and operational effectiveness.

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CONCLUSION

The Smart Campus Directory: AI-Powered Navigation System demonstrated significant improvements in campus navigation at the Universidad de Manila by integrating artificial intelligence, PWA technology, interactive kiosks, and QR code functionality. Through careful development and iterative prototyping, the system provided efficient, reliable, and user-friendly guidance that addressed persistent challenges caused by outdated maps and confusing layouts. User acceptance testing confirmed high usability and satisfaction, highlighting the

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system's positive impact on students, faculty, and visitors. Overall, the project not only enhances the daily campus experience but also sets a foundation for future innovations in smart educational environments.

REFERENCES

[1] M. Sedilla, R. Concepcion, J. Otilla, F. Lim, P. Velarde, and A. Samonte, "Artificial intelligence in navigational applications: Addressing GPS challenges in Metro Manila," Journal of Emerging Technologies in Navigation Systems, vol. 12, no. 3, pp. 45–56, 2024.
[2] J. Taruc and M. De La Cruz, "Application of Narrowband-IoT (NB-IoT) in smart campuses: A review of use cases and potentials," in Proc. Int. Conf. on Smart Education and IoT Systems, pp. 120–128, 2024.
[3] J. Oliva and R. Labicane, "Augmented reality campus guide for Romblon State University," Philippine Journal of Computing and Information Systems, vol. 10, no. 2, pp. 88–97, 2022.
[4] M. Archico, L. Santos, P. Dela Peña, and J. Rivera, "AR mobile application for campus navigation and information access at Pamantasan ng Lungsod ng Maynila," in Proc. IEEE Int. Conf. on Educational Technology and Smart Applications, pp. 201–210, 2024.
[5] R. Alegado, T. Mendoza, C. Bautista, and F. Ramirez, "NEUST Wayfinder: A GIS-based campus navigation system using Agile Development," International Journal of Information Systems and Smart Education, vol. 15, no. 1, pp. 34–42, 2024.
[6] E. Avelino, "Agile software development: An overview," International Journal of Computer Applications, vol. 975, no. 8887, pp. 23–28, 2018